Exercise Sheet 05

The Scriptorium Scroll

House rules.

- One file per exercise, in `my_work/`: `ex1_split_join.py`, `ex2_write_read.py`, …
- Run everything **from the lesson folder**: `python my_work/ex3_read_csv.py`. The data lives in `data/`, and the paths on this sheet (`data/…`, `my_work/…`) only work from there.
- Type the **Given** lines first, then write the rest. Compare with the **expected output** character by character.
- Files you create go into `my_work/`, never into `data/`. The scroll must survive you.
- Hand in: commit and push to `lesson-05`, open a pull request, and tell Lars which part of the Scriptorium's copy annoyed you most.

Exercise 1 Split and join

8 min · `ex1_split_join.py`

> **Raven from Orrin of the Clockworks:** *"The Scriptorium writes names as 'Surname, First'. And the supply tags are one long string with semicolons. Don't ask. Can you take them apart?"*

GIVEN

```
name_field = "Pennyworth, Nye"
tags = "quill;ink;parchment;parchment;ink"
sentence = "Counting the realm's coin since 1019"
```

(a) Split `name_field` into surname and first name, and print "First Last".

(b) Split the sentence into words. Print how many there are, then the words in reverse order, then the words joined with underscores, all lowercase.

(c) Split the tags, remove the duplicates (`set()`), sort them, and print them separated by ", ".

EXPECTED OUTPUT

```
Nye Pennyworth
6 words
1019 since coin realm's the Counting
counting_the_realm's_coin_since_1019
ink, parchment, quill
```

Hint: `last, first = name_field.split(", ")` unpacks both halves in one go. `reversed(words)` and `sorted(set(tag_list))` each give you something `" ".join()` can eat.

Exercise 2 Write a file, read it back

10 min · `ex2_write_read.py`

> **Decree from Queen Aldara:** *"I keep the Scriptorium's tallies on a scrap of vellum. Orrin says a 'text file' is more robust. Prove it."*

GIVEN

```
lines = ["Scrolls copied: 1019", "Scrolls smudged: 17", "Ink pots used: 203"]
```

(a) Write the three lines into `my_work/scriptorium_note.txt`, one per line. Use `with open(..., "w")` and remember the `"\n"`.

(b) Open the file again, read the whole thing into one string with `.read()`, print it, and print how many characters and lines it has.

(c) Open it a third time and loop over it line by line. Split each line at `": "` and print the label left-aligned in 20 characters and the number right-aligned in 6.

EXPECTED OUTPUT

```
Scrolls copied: 1019
Scrolls smudged: 17
Ink pots used: 203
Characters: 60
Lines: 3
Scrolls copied        1019
Scrolls smudged         17
Ink pots used          203
```

Hint: Each line you read still ends in `"\n"`. `.strip()` it before splitting, or the number won't line up. `print(text, end="")` stops print from adding a fourth newline.

Exercise 3 Open the scroll

8 min · `ex3_read_csv.py`

> **Raven from Mistress Pell (Paymaster):** *"The Scriptorium's copy of the Roll is in `data/scriptorium_export_messy.csv`. Don't open it in Excel, Excel 'helps' with the dates. Just tell me what's in it."*

GIVEN

```
import csv
```

(a) Open the file with `csv.DictReader` and print the column names (`reader.fieldnames`).

(b) Put all rows in a list. Print how many there are, then for the first three rows print the ref, the name and the salary, separated by " | ".

(c) Print the first row's salary with `repr()` so you can see exactly what text arrived.

EXPECTED OUTPUT

```
Columns: ['Employee Ref', 'Name', 'Guild', 'Hours', 'Salary', 'Bonus', 'Gender', 'Start']
Rows: 40
 TM1092  | Pennyworth, Nye | 41,900 cr
tm1178 | Woolpack, Hild | 26550
tm1105 | MARCHBANK, MAREN | 24950
First salary as text: '41,900 cr'
```

Hint: Always open CSV files with `newline=""` and `encoding="utf-8"`, or odd characters and line endings go wrong on Windows. Yes, the spaces around `TM1092` are really in the file. The Scriptorium copies by hand.

Exercise 4 Clean it up

12 min · `ex4_clean.py`

> **Raven from Mistress Pell:** *"Refs in three different styles, names SHOUTED or whispered, salaries with 'cr' or without, hours with 'hrs' on the end. Every. Single. Moon."*

GIVEN

```
import csv
```

(a) Write three functions: `clean_ref(text)` (strip, uppercase, add "TM" if it's missing), `clean_name(text)` (split at the comma, strip both halves, Title Case, "First Last") and `clean_money(text)` (strip, remove "cr" and commas, empty text means `0.0`, otherwise a float).

(b) Read the scroll and, for the first 8 rows, print the clean ref, the name left-aligned in 20, the hours as a number right-aligned in 6, and the salary as crowns right-aligned in 13.

EXPECTED OUTPUT

```
TM1092  Nye Pennyworth        37.5   41,900.00 cr
TM1178  Hild Woolpack         37.5   26,550.00 cr
TM1105  Maren Marchbank       37.5   24,950.00 cr
TM1180  Una Wildmoor          37.5   29,000.00 cr
TM1189  Nell Oakhollow        37.5   31,750.00 cr
TM1082  Jory Eastmarch        37.5   37,650.00 cr
TM1037  Ulric Wildmoor        37.5   25,900.00 cr
TM1199  Cadoc Nightingale     22.5   37,950.00 cr
```

Hint: Hours: `float(row["Hours"].replace("hrs", "").strip())`. One name has spaces *inside* the comma ("Fairweather , Maddox"), so strip *after* splitting. Make the crowns text first (`f"{salary:,.2f} cr"`), then right-align it.

Exercise 5 Write the clean file

12 min · `ex5_write_csv.py`

> **Raven from Lady Siv of the Rewards:** *"Lovely. Now can I have that as a file I can actually open? Columns: employee_id, name, gender, hours, salary, bonus. Gender as one letter, M or F, whatever the scribe felt like that day."*

GIVEN

```
import csv
```

(a) Reuse your three cleaning functions (copy them in) and add `clean_gender(text)`: the first letter, uppercase.

(b) Build a list of dicts, one per row, with exactly the six keys Lady Siv asked for.

(c) Write it to `my_work/roll_clean.csv` with `csv.DictWriter`, header included. Print how many rows you wrote.

(d) Open your new file as plain text and print its first four lines, to prove it's real.

(e) Count the women in the scroll and print the number.

EXPECTED OUTPUT

```
Wrote 40 rows
employee_id,name,gender,hours,salary,bonus
TM1092,Nye Pennyworth,M,37.5,41900.0,0.0
TM1178,Hild Woolpack,F,37.5,26550.0,0.0
TM1105,Maren Marchbank,F,37.5,24950.0,0.0
Women in the scroll: 15
```

Hint: `fieldnames=list(clean_rows[0])` uses the keys of the first dict as the header. Open the output with `newline=""` too, or Windows gives you a blank line after every row.

Exercise 6 Guild totals

10 min · `ex6_department_totals.py`

> **Decree from Queen Aldara:** *"The Privy Council asked which guild costs the realm the most. I said 'The Forge, probably?' They wrote that down. Please make it true or false by the next council."*

GIVEN

```
import csv
```

(a) Read `data/thornmere_payroll.csv` (the clean Roll of the Realm, 200 subjects) and build two dicts: total salary per guild and headcount per guild. The guild is in the `department` column.

(b) Print a table, guilds sorted alphabetically: name in 12, people in 6, total in 14 with thousands separators, average in 12 with no decimals.

(c) Print the guild with the biggest wage bill.

EXPECTED OUTPUT

```
Guild       People         Total     Average
Apothecary      15       495,250      33,017
Clockworks      20       808,300      40,415
Forge           60     1,748,900      29,148
Guard           35     1,129,250      32,264
Harbour         25       771,100      30,844
Kitchens        12       397,550      33,129
Library         15       463,650      30,910
Treasury        18       585,850      32,547
Biggest wage bill: Forge
```

Hint: `totals.get(dept, 0) + salary` works whether the guild is already in the dict or not. `max(totals, key=totals.get)` gives the key with the largest value.

Exercise 7 Files about files STRETCH

12 min · `ex7_pathlib_report.py`

> **Raven from Orrin:** *"Which files are even in that data folder? And can the answer be a file too, so I can send it to the Queen by raven instead of explaining it?"*

GIVEN

```
import csv
from pathlib import Path
```

(a) Use `Path("data").glob("*.csv")` to find every CSV in the data folder. For each one (sorted), print its name left-aligned in 28 and how many data rows it has (lines minus the header) right-aligned in 5.

(b) Find the highest-paid subject in the Roll.

(c) Write a three-line report to `my_work/report.txt`: a title, "Highest paid: First Last (Guild)", and the salary in crowns. Then print "Wrote report.txt with 3 lines" and the report itself.

EXPECTED OUTPUT

```
scriptorium_export_messy.csv   40 rows
thornmere_payroll.csv         200 rows
Wrote report.txt with 3 lines
ROLL OF THE REALM: QUICK FACTS
Highest paid: Linnet Quickwater (Treasury)
Salary: 68,750 cr
```

Hint: A `Path` can be opened like a filename, and `Path("my_work") / "report.txt"` builds a path with a slash. `report.read_text()` reads a whole file in one call. Print `report.name`, not `report`: Windows shows paths with backslashes.

Bug hunt Pages from the Grimoire, page 5

15 min · `bughunt/export_bug.py`

Dalvric's monthly tally of the Roll. Mistress Pell says it "worked in his tower". There are **three** bugs: one crashes before a single row is read, one gives a wrong count without crashing, and one loses the pennies.

EXPECTED OUTPUT ONCE FIXED

```
Subjects: 200
Full-time: 174
Total salary: 6,399,850 cr
Average salary: 31,999.25 cr
```

Commit each fix separately, with a message explaining what was wrong.

More riddles. Five more torn pages wait in `bughunt/riddles/`, one bug each (the last one has two), getting harder as you go. Optional: for fast finishers, or a quiet evening.